

Web Programming Final Exam Questions and Answers

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Academic Year 2024-2025

Question 1

What is the DOM and what is its application? (1 point)

Answer:

The DOM (Document Object Model) is a tree-like structure of HTML code created by the browser. Using this tree structure, a programming language (such as JavaScript) can access page elements (tags) and perform modifications such as changing text, styles, adding or removing tags, and controlling events.

Question 2

What architecture does Django use? Explain its components completely. (3 points)

Answer:

Django uses the **MVT (Model-View-Template)** architecture. Its components are:

- **Model (M):** Used to define data structures and database interaction, such as tables for users, shopping carts, and articles.
- **Template (T):** The presentation layer or HTML code that uses special Django tags to display dynamic data to the user.
- **View (V):** The application logic that receives user requests, processes them (interacting with the model), and returns the final output to the Template.

Question 3

Suppose you want to apply two types of discounts to a user visiting the site. Explain the implementation of each. (2 points)

- a) Randomly generate a discount code on the client side and display it to the visitor. The code should not change when the page is refreshed.
- b) If the user has made more than two purchases from the store in the past month, apply a discount code equal to three percent of their total purchases in the past month.

Answer:

a) First, we check using JavaScript code whether a discount code is already stored in `localStorage` or not. If it exists, we display it via `window.alert` (or other methods). If it does not exist, we generate a random number or string (using `Math.random`), store it in `localStorage`, and then display it. This ensures that refreshing the page does not change the code.

b) This part must be implemented on the backend side because it requires the user's purchase history. In the database (orders table), the user's orders within the past month interval are filtered, and their count is calculated (`count`). If this count is greater than 2, their purchase amounts in the past month are summed up, multiplied by 0.03 (three percent), and the resulting amount is deducted from the current invoice.

Question 4

Examine the HTML output of the given table in the sheet.

a) Draw the output of the code.

b) If the `rowspan="2"` attribute is removed from the Image column, how does the table shape change?

```
<table border="1">
<tr><th rowspan="2">Img</th><th rowspan="2">Name</th><th colspan="2">Contact</th></tr>
<tr><th>Email</th><th>Phone</th></tr>
<tr><td>IMG</td><td>Ali</td><td>a@x.com</td><td>0912</td></tr>
<tr><td>IMG</td><td>Reza</td><td colspan="2">No Contact</td></tr>
</table>
```

Answer:

a) Drawing the code output:

Img	Name	Contact	
		Email	Phone
IMG	Ali	a@x.com	0912
IMG	Reza	No Contact	

b) **Effect of removing `rowspan="2"`:** If this property is removed, the `Img` cell no longer spans two rows. As a result, a row in the table will become empty and the structure will get distorted, such that in the second row of headers, the word `Email` shifts under the `Img` column and the cells shift.

Question 5

Answer the following questions based on the given CSS and JS codes in the question.
(3 points)

```
<style>
html{ font-size:10px; }
#box{ color: blue; font-size:2rem; }
.item{ color: green; }
.container .item{ color: orange; }
.container > .item.active{ color: red; }
.card{ width:10vw; height:5vh; display:flex;
      justify-content:center;
      align-items:center; }
</style>
</head>

<body>
<div class="container">
  <p id="box" class="item active">Text</p>
</div>
<div class="card">Card</div>

<script>
$(".card").click(function(){
  $("#box").css("color","purple");
});
</script>
```

Answer:

- **a)** The text color of element p is **blue** (since the ID selector or #box has the highest priority).
- **b)** The font size of <p> is **20 pixels** (the root value is 10px, so 2rem equals 20 pixels).
- **c)** The vw unit in width: 10vw is calculated relative to the **Viewport Width**.
- **d)** The selector #box is an **ID Selector**.
- **e) Script code functionality:** This code is written using the jQuery library. Its functionality is that by clicking on any element with the class card, the text color of the element whose ID is box changes to purple.

Question 6

In the given JavaScript code, specify the requested items (3.5 points)

```
function outer() {  
  let x = 5;  
  return function inner(y) {  
    return x + y;  
  };  
}  
let fn = outer();  
  
let arr = [1,2,3];  
function pushToArr(array, val){  
  array.push(val);  
}  
  
const multiply = (n) => n*2;  
const anon = function(n){ return n+3; }  
function run(fn, val){ return fn(val); }  
  
let num = 5;  
let str = "5";  
  
pushToArr(arr, 4);  
let r1 = fn(10);  
let r2 = run(multiply, 3);  
let r3 = run(anon, num);  
let r4 = (num === str);
```

Answer:

- a) **Value of r1:** Equal to **15** (the function `outer` keeps the value $x = 5$ in its closure, and `inner(10)` returns $5 + 10$).
- b) **Value of r2:** Equal to **6** (the function `run` passes 3 to the function `multiply` and the result becomes $3 \times 2 = 6$).
- c) **Value of r3:** Equal to **8** (the function `run` passes `num`, which is 5, to `anon` and the result becomes $5 + 3 = 8$).
- d) **Value of r4:** Equal to **false** (because 5 as a number and "5" as a string are not equal and not of the same type under `===`).
- e) **Callback functions:** `multiply` and `anon`
- f) **Function expressions:** `multiply` and `anon`
- g) **Function declarations:** `outer`, `pushToArr`, and `run`

Question 7

Answer the questions based on the Django Python code: (1.5 points)

```
1  # models.py
2  class Article(models.Model):
3      title = models.CharField(max_length=200)
4      slug = models.SlugField(unique=True)
5      body = models.TextField()
6      author = models.ForeignKey(
7          User,
8          on_delete=models._____,
9          related_name='articles'
10     )
11     created_at = models.DateTimeField(auto_now_add=True)
12
13     def __str__(self):
14         return _____
15
16 # admin.py
17 @admin.register(Article)
18 class ArticleAdmin(admin.ModelAdmin):
19     list_display = ('title', 'author', 'created_at')
```

Answer:

- **a)** To delete articles when the author is deleted (line 8), the following command should be used: **CASCADE** (more precisely: `models.CASCADE`)
- **b)** To display the article title in the admin panel (in the `__str__` method), the blank in line 14 should be filled as: **`self.title`**
- **c) Role of `related_name`:** This feature enables **Reverse Relation**; meaning that through the `User` model (author), all of their articles can be accessed, for instance via the command `author.articles.all()`.